Predictive Analytics for Workforce Stability: Identifying Attrition Factors with Supervised Machine Learning

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Abstract

High employee turnover poses significant financial and organizational challenges across industries. This project applies an end-to-end data science approach to predict employee attrition using supervised machine learning models. Using the IBM Watson HR Employee Attrition and Performance synthetic dataset of 1,470 employee records with 35 features, the project encompasses data acquisition, preprocessing, exploratory data analysis (EDA), model development, and evaluation using Python. Two algorithms were compared: Logistic Regression and Random Forest. The results show that Logistic Regression achieved a higher Area Under the Curve (AUC) of 0.8374 on the test set, demonstrating stronger discriminative ability for identifying employees at risk of leaving, while Random Forest achieved higher raw accuracy at 84.69%. Key drivers of attrition identified include monthly income, age, overtime status, total working years, and years at the company. The results aim to support proactive, data-driven retention strategies while acknowledging limitations inherent in synthetic historical data.

I. Introduction

Employee attrition is the gradual reduction of a workforce as employees leave an organization and are not immediately replaced. It can be voluntary, such as when an employee resigns, or involuntary, such as in cases of layoffs or terminations. Regardless of the cause, attrition creates real costs for organizations, including recruiting and onboarding new employees, lost institutional knowledge, reduced team morale, and decreased productivity during reduced headcount periods. Research shows that replacing an employee can cost between 1.5 and 2 times that employee's annual salary, which can amount to millions of dollars for large organizations [1].

Beyond financial costs, attrition affects whole industries and economies. When skilled workers leave at high rates, companies lose competitive advantages and entire sectors can experience talent shortages. For example, in healthcare, high nurse turnover can directly affect patient outcomes. In technology, losing experienced engineers slows product development. Understanding why employees leave — and predicting who is likely to leave before they do — is one of the most valuable problems that data science can help solve.

Having personally witnessed high employee turnover across multiple roles, the motivation behind this project comes from seeing first-hand how reactive retention strategies often fail. Most organizations only respond to attrition after it happens, rather than identifying warning signs early. This practicum project addresses that gap by building machine learning models that can flag employees at elevated risk of leaving, giving organizations the opportunity to intervene before the departure occurs.

Unlike typical curriculum assignments that isolate individual stages of the data lifecycle, this project integrates every stage — from data collection and preparation through exploratory analysis, modeling, and evaluation — into one cohesive, real-world workflow. This end-to-end approach reflects the complexity of professional data science practice and serves as a capstone demonstration of the skills developed throughout the Master of Science in Data Science program.

II. Problem Statement

The central question guiding this project is: *How can machine learning models be effectively developed and applied to predict employee attrition?* This question addresses a genuine need for data science approaches because simpler solutions — such as exit surveys or manager intuition — fail to capture the nuanced, multi-variable drivers that lead employees to leave.

This project engages with the full data lifecycle. It begins with data collection using a synthetic HR dataset capturing a wide range of employee-related variables including demographic information, job characteristics, satisfaction ratings, and performance data. Data preparation includes cleaning, feature engineering, and transforming skewed variables. Exploratory data analysis (EDA) uses descriptive statistics and visualizations to uncover initial patterns and relationships. The modeling phase involves training and evaluating multiple supervised machine learning algorithms. Finally, results are visualized and compiled into this report with actionable recommendations.

Employee turnover can cost companies 1.5 to 2 times an employee's annual salary in recruitment, training, and lost productivity, potentially amounting to millions for large organizations [1]. By predicting attrition, organizations can proactively intervene, saving substantial financial resources and preserving institutional knowledge.

III. Related Work

Govindarajan et al. [2] conducted a comparative analysis of machine learning models for predicting attrition using the same IBM HR Analytics dataset used in this project. Their study evaluated several algorithms — including Decision Trees and Random Forests — and found that ensemble methods generally outperformed simpler classifiers in terms of accuracy. However, their work focused primarily on overall accuracy as the main performance metric, which can be misleading when dealing with imbalanced datasets where the minority class (past employees) is significantly underrepresented. This project addresses that limitation by also evaluating AUC, precision, recall, and F1-score to provide a more complete performance picture.

Manafi Varkiani et al. [1] investigated attrition prediction with a focus on identifying and explaining the key determinants of turnover. Their findings highlighted the importance of interpretability in HR analytics — it is not enough to predict who will leave. Organizations also need to understand why, and this insight directly motivates the feature importance analysis performed in this project using the Random Forest model.

Couronné et al. [3] performed a large-scale benchmark comparing Random Forest and Logistic Regression across more than 250 classification datasets. Their results showed that Random Forest tends to outperform Logistic Regression in accuracy, but Logistic Regression maintains advantages in interpretability and AUC performance in certain contexts, particularly with class imbalance present. This research directly supports the decision to compare both algorithms in this project.

Hoffman and Tadelis [4] examined the role of people management quality in employee attrition, finding that poor managerial performance was significantly associated with higher turnover rates among subordinates. Their work emphasizes that attrition is not driven solely by compensation — the quality of the manager-employee relationship is a powerful independent factor, which is reflected in this project's inclusion of variables such as years with current manager and relationship satisfaction.

IV. Data Description

A. Dataset Overview

The dataset used in this project is the IBM Watson HR Employee Attrition and Performance dataset, a synthetic dataset made publicly available on Kaggle. It contains 1,470 employee records and 35 features covering a wide range of employee attributes, including demographic variables such as age, gender, marital status, and distance from home; compensation-related variables such as monthly income and hourly rate;

job characteristics such as job role, department, and job level; satisfaction and engagement metrics such as job satisfaction and environment satisfaction; and career history variables such as total working years, years at the company, and years with current manager.

The target variable is "Attrition," a binary indicator of whether an employee left the company (Yes) or remained (No). Of the 1,470 records, 1,233 employees (83.88%) were still with the company at the time of data collection, while 237 employees (16.12%) had left. This class imbalance is important to account for in the modeling process, as it can cause models to be biased toward predicting the majority class.

B. Data Ethics and Governance

Real employee data is difficult to obtain for research purposes. Companies do not make personnel records public, and accessing non-synthetic demographic data typically requires non-disclosure agreements, making it inaccessible for open academic research. The use of a synthetic dataset addresses these concerns while still preserving the statistical properties needed for meaningful analysis [5].

The use of synthetic data itself raises ethical considerations worth acknowledging. Susser et al. [6] note that synthetic health and organizational data carries both promise and peril — while it removes privacy risks associated with real identifiable individuals, it can introduce its own biases depending on how the synthetic data was generated. Additionally, machine learning models trained on synthetic data may not generalize perfectly to real organizational environments. Cooper and Coetzee [7] further emphasize that even with publicly available data, researchers must consider purpose, potential harm, and transparency in how results are communicated.

A well-known example of re-identification risk from real data is the Netflix Prize dataset challenge, in which researchers were able to reverse-engineer anonymized customer records to identify individuals [8]. This case reinforces why synthetic data is the responsible choice for this type of research when real employee records are not available through proper consent channels. All data used in this project was handled locally with no personally identifiable information involved.

V. Methodology

A. Tools and Libraries

The entire project was implemented in Python. Core libraries used include `pandas` and `NumPy` for data manipulation, `Matplotlib` and `Seaborn` for static data visualization, `Plotly Express` for interactive visualization, `SciPy` for statistical testing, and `scikit-learn` for machine learning modeling, preprocessing, and evaluation.

B. Preprocessing Steps

Before modeling, the dataset underwent a series of preprocessing steps. Four columns were removed due to providing no analytical value: `EmployeeCount` (constant value of 1), `StandardHours` (constant at 80), `Over18` (constant "Y"), and `EmployeeNumber` (a non-informative ID field). Variables representing ordered categorical ratings — such as job satisfaction, environment satisfaction, work-life balance, job involvement, job level, and performance rating — were cast as categorical types, along with all nominal string columns such as department, job role, gender, marital status, business travel, and overtime.

A skewness analysis was then conducted on all remaining numerical features. Columns with an absolute skewness value greater than 0.75 were log-transformed using $\log(x+1)$, which compresses extreme values and brings distributions closer to normality. Nine features met this threshold: `DistanceFromHome`, `MonthlyIncome`, `NumCompaniesWorked`, `PercentSalaryHike`, `TotalWorkingYears`, `YearsAtComp` `YearsInCurrentRole`, `YearsSinceLastPromotion`, and `YearsWithCurrManager`.

C. Statistical Testing

Prior to modeling, an independent samples *t*-test was conducted to assess whether the difference in average monthly income between employees who left and those who stayed was statistically significant. Levene's test first confirmed equal variances ($p = 0.7586$). The *t*-test returned a test statistic of $t = -7.7477$ with a p-value effectively equal to zero ($p < 0.0001$), leading to the rejection of the null hypothesis. Past employees earned significantly lower average monthly incomes than current employees, confirming that income is a meaningful predictor of attrition.

D. Model Training and Evaluation

The dataset was split into training (60%), validation (20%), and test (20%) sets using stratified sampling to preserve the class distribution across all three sets. All categorical features were one-hot encoded using `pandas get_dummies` with `drop_first=True` to avoid multicollinearity. Boolean columns were converted to integers and all features were standardized using `StandardScaler`.

Two classification models were trained and evaluated. Logistic Regression was implemented with a balanced class weight setting to counteract the class imbalance, using the LBFGS solver with a maximum of 1,000 iterations. Random Forest was tuned using `RandomizedSearchCV` with 50 iterations and 5-fold cross-validation, optimizing for AUC as the scoring metric. The best hyperparameters found were 336 estimators, `log2` max features, a minimum of 8 samples per leaf, and a minimum of 17 samples to split an internal node. Models were evaluated using accuracy, AUC, precision, recall, F1-score, and average precision.

VI. Exploratory Data Analysis

A. Attrition Overview

Of the 1,470 employees in the dataset, 1,233 (83.88%) were still employed while 237 (16.12%) had left. This class imbalance is a common challenge in attrition data, as models can appear accurate simply by always predicting the majority class. Balanced class weights and AUC-focused tuning were applied to both models to address this issue.

B. Attrition by Department

Sales showed the highest attrition rate relative to its size, followed by Human Resources, while Research and Development had comparatively lower turnover. These differences may reflect variations in work demands, management culture, or compensation structures across departments. Figure 1 illustrates the distribution of current and past employees across departments.

C. Attrition by Age

Younger employees in their mid-to-late twenties and early thirties accounted for the largest share of departures, with attrition dropping off noticeably after age 35. This aligns with early-career job mobility patterns, where younger workers are more likely to change positions before settling into long-term roles [4]. Figure 2 shows attrition counts across the age spectrum.

D. Attrition by Compensation

Hourly rate showed no clear directional relationship with attrition. Monthly income, however, told a different story — employees who left were concentrated in lower income brackets ($1,000–$4,000), while current employees were more evenly distributed across higher ranges. This pattern was confirmed by the *t*-test, which found a statistically significant difference in mean monthly income between the two groups ($p < 0.0001$). Figure 3 displays attrition counts by monthly income.

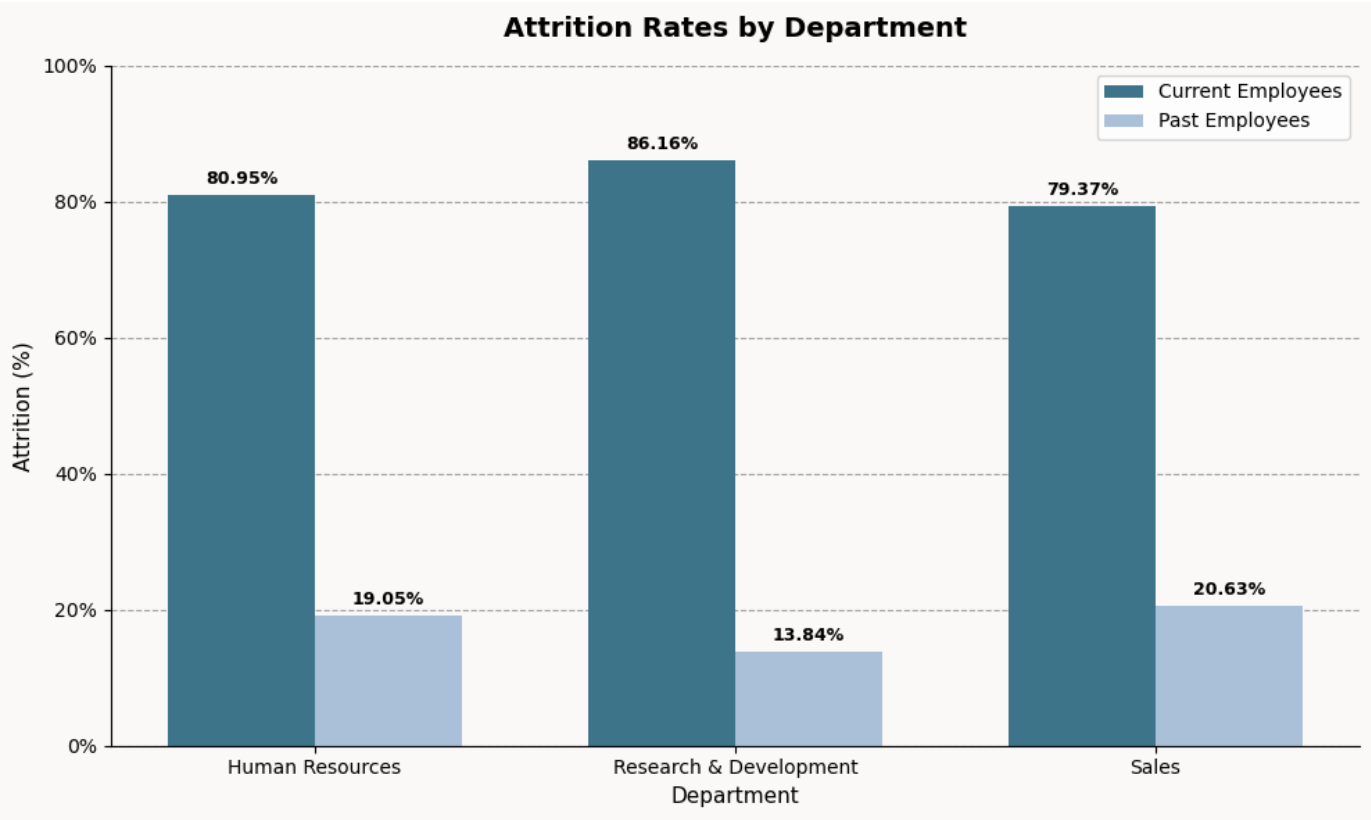

Fig. 1. Attrition rates by department.

E. Attrition by Years with Manager

Employees who left were most concentrated among those with 0 to 2 years with their current manager, suggesting that newer or recently reassigned manager-employee relationships are more vulnerable to turnover. Hoffman and Tadelis [4] found that manager quality and relationship tenure are significant factors in attrition decisions. Figure 4 shows attrition counts by years worked with the current manager.

F. Correlation Matrix

The correlation matrix highlighted several key relationships. Monthly income correlated positively with job level and total working years. Attrition correlated negatively with job level, monthly income, total working years, and stock option level, confirming that more senior, better-compensated, longer-tenured employees are less likely to leave. Overtime showed a positive correlation with attrition, reinforcing its role as a key risk factor. These patterns directly guided feature selection and model interpretation. Figure 5 displays the full correlation matrix.

VII. Model Results and Comparison

A. Logistic Regression

The Logistic Regression model, trained with balanced class weights to address the attrition class imbalance, achieved an accuracy of 78.91% on the validation set and 79.25% on the test set. More importantly, it achieved AUC scores of 0.8413 (validation) and 0.8374 (test), indicating strong discriminative ability between employees who left and those who stayed. The recall for the minority class (past employees) was 0.6875 on the validation set and 0.7660 on the test set, meaning the model correctly identified approximately 69–77% of employees who had actually left — a critical metric for a retention-focused application where missing a high-risk employee is costly. The F1-score for the minority class was 0.5156 (validation) and 0.5414 (test).

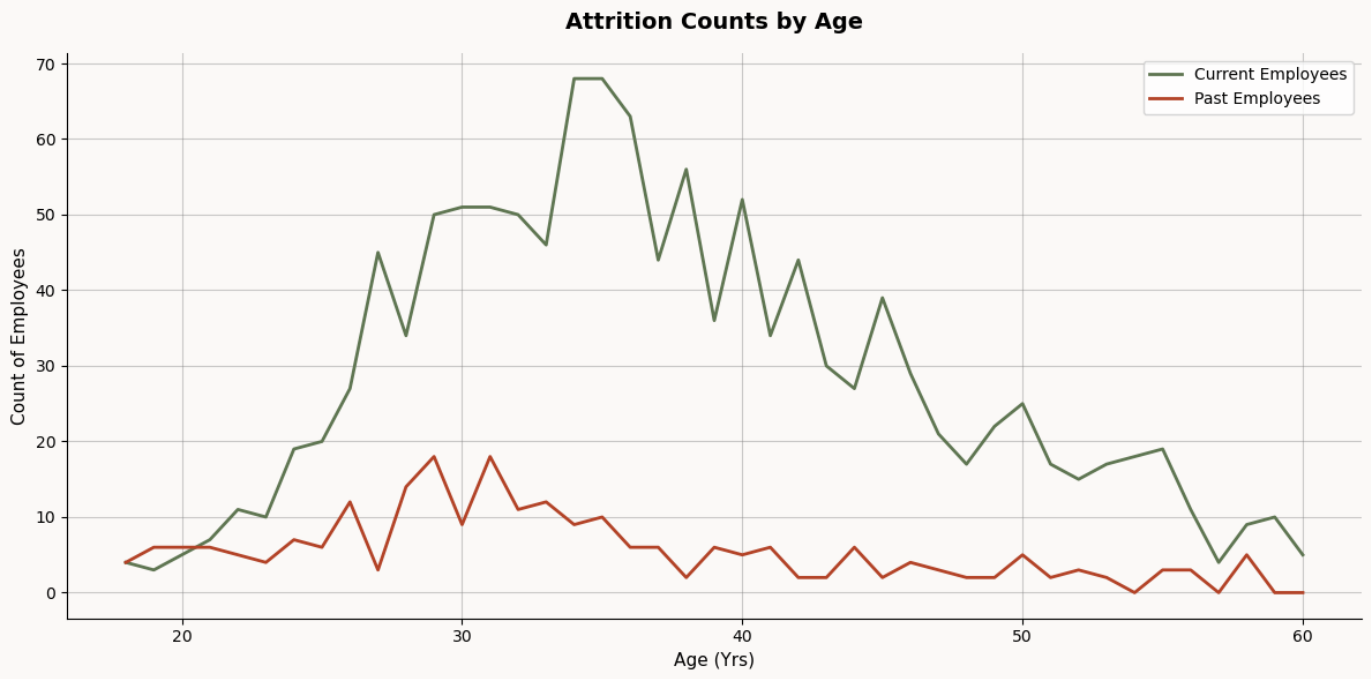

Fig. 2. Attrition counts by employee age.

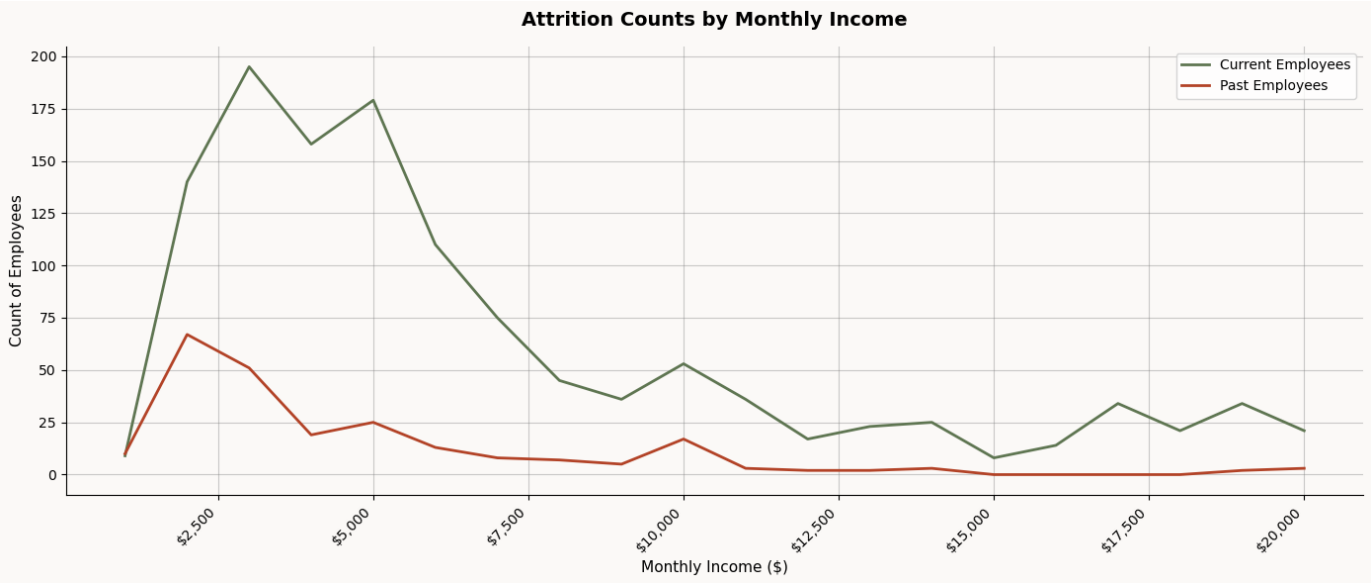

Fig. 3. Attrition counts by monthly income.

B. *Random Forest*

The Random Forest model, tuned via `RandomizedSearchCV`, achieved higher raw accuracy than Logistic Regression: 83.33% on the validation set and 84.69% on the test set. However, this accuracy came at a significant cost to recall for the minority class. On the validation set, the Random Forest correctly identified only 2.08% of employees who had left (recall $= 0.0208$), and on the test set, only 4.26% (recall $= 0.0426$). The model almost entirely defaulted to predicting "current employee" for most records, artificially inflating its accuracy while providing almost no practical value for identifying who is at risk of leaving. AUC scores of 0.7952 (validation) and 0.7719 (test) were also lower than those achieved by Logistic Regression.

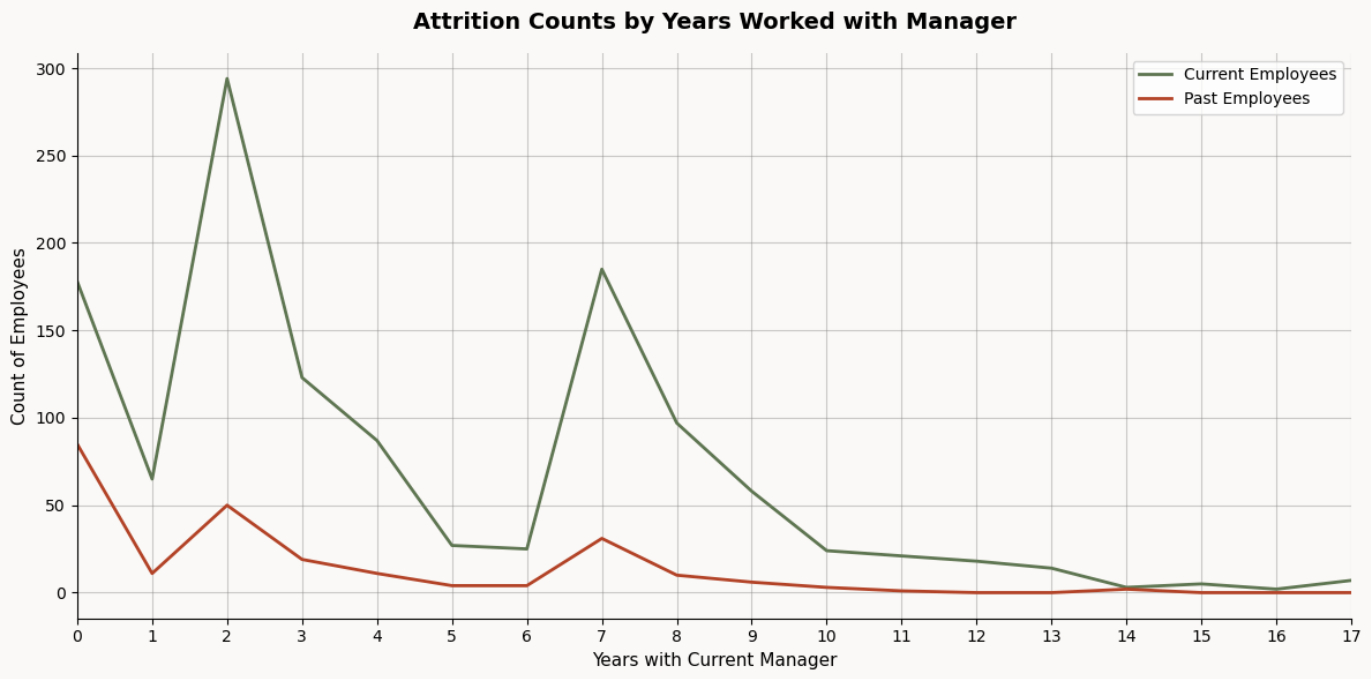

Fig. 4. Attrition counts by years with current manager.

C. Model Comparison Summary

Table I summarizes all evaluation metrics for both models across the validation and test sets.

TABLE I
MODEL COMPARISON — METRIC SUMMARY

Model	Acc.	AUC	Prec.	Recall	F1	Avg P
RF — Val	83.33%	0.7952	0.3333	0.0208	0.0392	0.4285
RF — Test	84.69%	0.7719	1.0000	0.0426	0.0816	0.5128
LR — Val	78.91%	0.8413	0.4125	0.6875	0.5156	0.5824
LR — Test	79.25%	0.8374	0.4186	0.7660	0.5414	0.6587

The precision-recall curves further illustrate the difference between the two models. Logistic Regression maintained a stronger balance between precision and recall across different classification thresholds, with average precision scores of 0.5824 (validation) and 0.6587 (test). Random Forest showed better precision at very high-confidence predictions but collapsed rapidly in recall, producing average precision scores of 0.4285 (validation) and 0.5128 (test). All curves significantly outperformed the random baseline of approximately 0.1612. Figure 6 shows the precision-recall curves for both models.

D. Feature Importance

The Random Forest model provides feature importance scores indicating which variables contributed most to its predictions. Figure 7 shows the top 15 most important features.

`MonthlyIncome` ranked highest at 0.1041, followed by `Age` (0.0766), `OverTime_Yes` (0.0740), `TotalWorkingYears` (0.0676), and `YearsAtCompany` (0.0523). These findings align with prior research. Manafi Varkiani et al. [1] similarly found income, seniority, and overtime to be among the strongest determinants of attrition. The role of `MaritalStatus_Single` reflects findings suggesting that single employees, with fewer location-based or family-based constraints, are more mobile in their career choices [9].

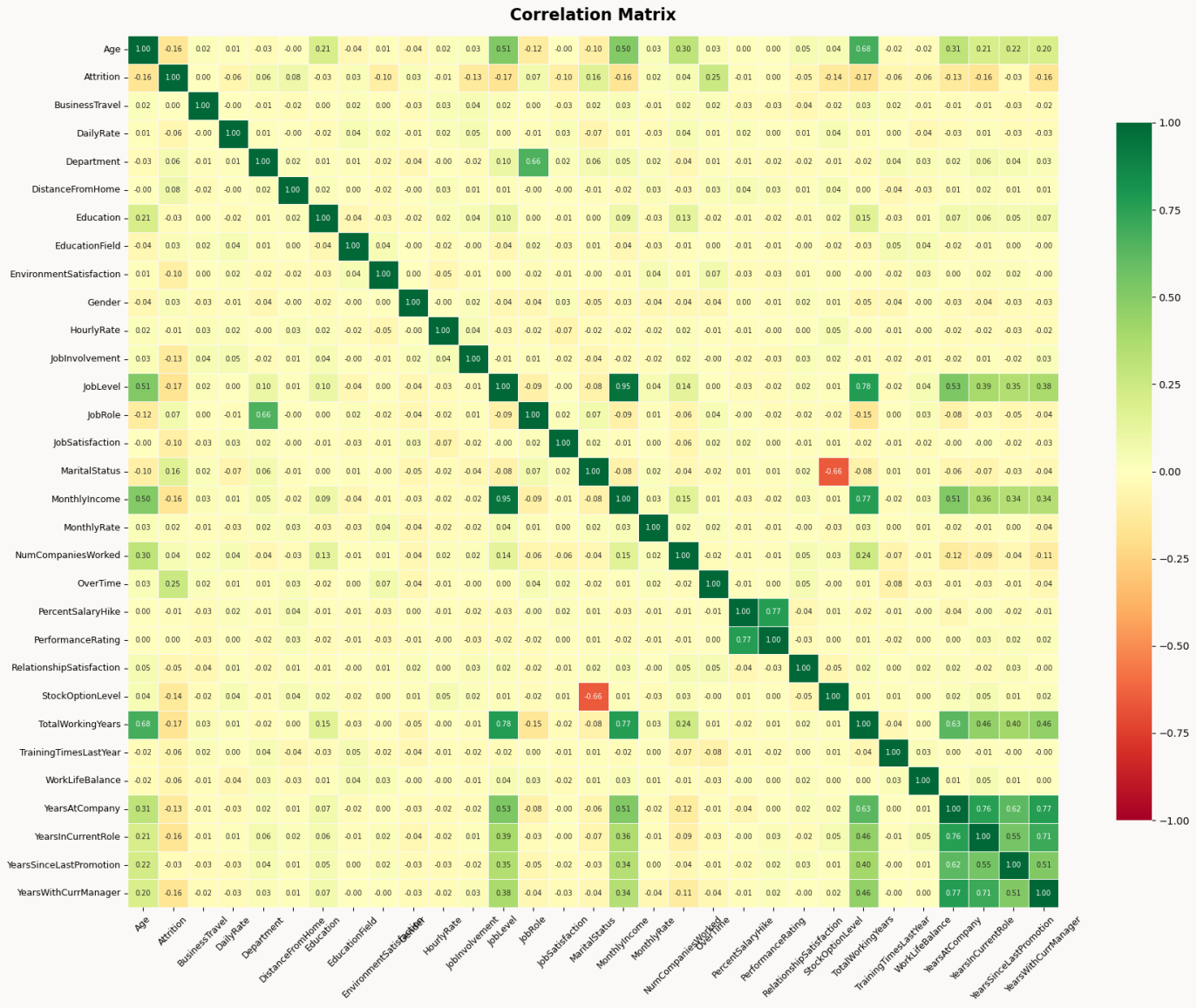

Fig. 5. Correlation matrix across all numeric and encoded variables.

VIII. Discussion and Key Findings

Compensation is a statistically confirmed driver of attrition. The *t*-test showed that employees who left earned significantly lower monthly incomes than those who stayed, and `MonthlyIncome` was the single most important feature in the Random Forest model. Organizations should regularly benchmark pay against market rates to reduce turnover risk.

Overtime is a major risk factor. Employees required to work overtime were substantially more likely to leave, suggesting issues with workload management, understaffing, or a culture that does not respect work-life balance.

Tenure and career stage matter. Younger employees and those early in their tenure at the company, in their role, or with their manager showed higher attrition rates, pointing to the value of structured onboarding and proactive manager support in the first two years.

Model selection must match the problem. Logistic Regression correctly identified 77% of at-risk employees on the test set, compared to just 4% for Random Forest, making it the clearly superior choice for this use case. For imbalanced attrition data, recall and AUC matter more than raw accuracy, a finding

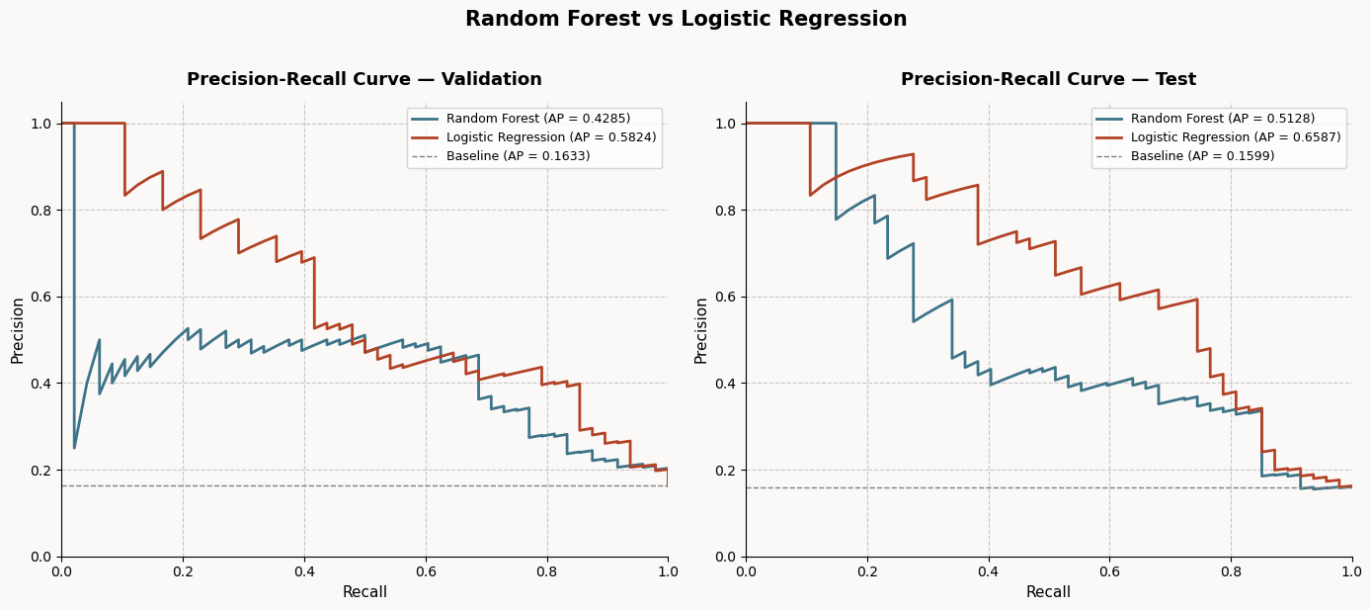

Fig. 6. Precision-recall curves: Random Forest vs. Logistic Regression on validation and test sets.

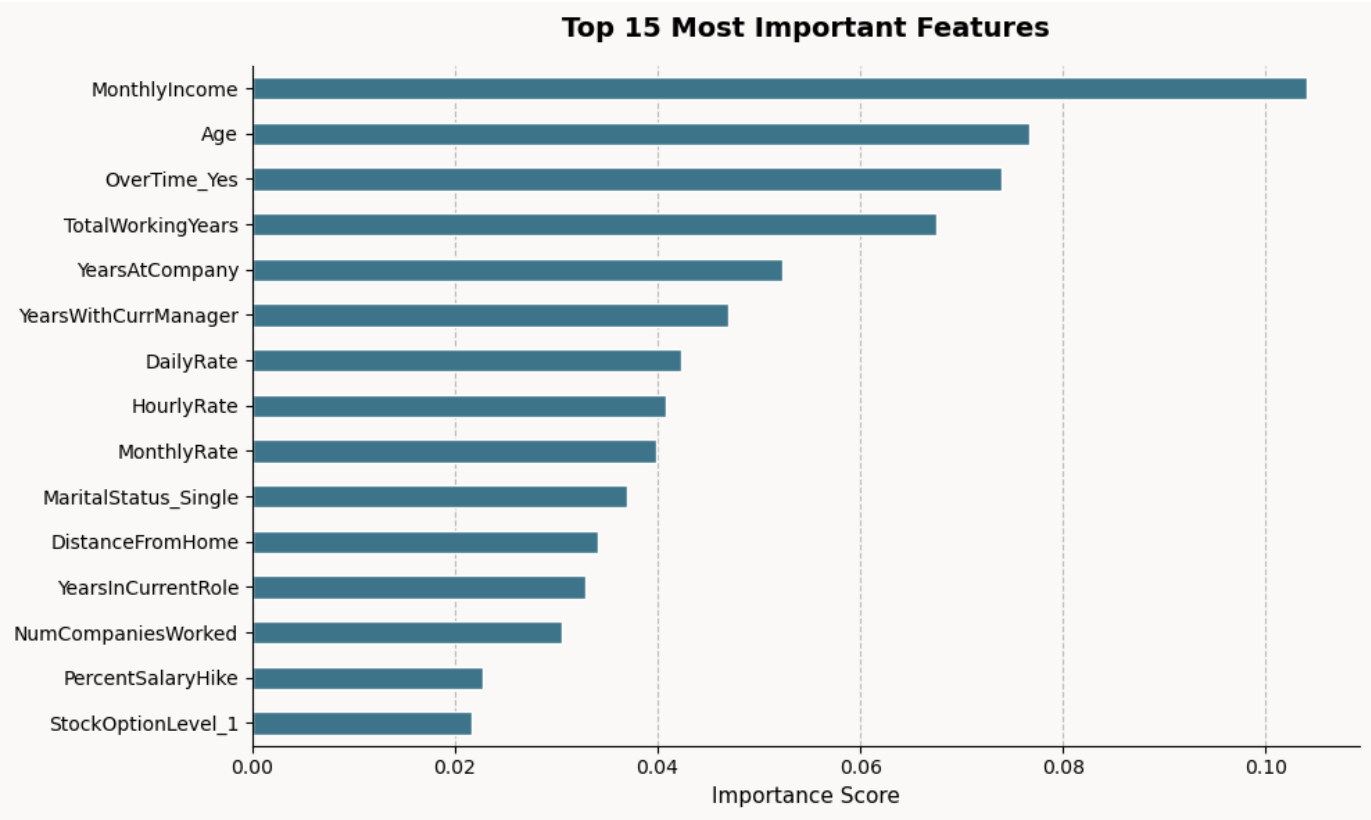

Fig. 7. Top 15 most important features from the Random Forest model.

consistent with Kirasich et al. [10]. Future work could explore incorporating unstructured text data, as demonstrated by Lee and Algarra [11].

IX. Timeline

The project was completed over an eight-week period structured as follows:

TABLE II
PROJECT TIMELINE

Duration	Phase	Description
Weeks 2–3	Project Setup & Data Acquisition	Defined project scope, objectives, and success metrics. Identified and acquired the IBM HR dataset. Conducted initial data review for suitability and quality.
Weeks 3–4	Preprocessing & EDA	Performed data cleaning, dropped zero-variance columns, encoded categorical variables, applied log transformations for skewed features, and conducted exploratory data analysis including visualizations and statistical testing.
Weeks 4–5	Model Development	Split data into training, validation, and test sets. Implemented Logistic Regression with balanced class weights. Implemented Random Forest with hyperparameter tuning via `RandomizedSearchCV`.
Weeks 5–6	Model Evaluation & Interpretation	Compared model performance across accuracy, AUC, precision, recall, and F1-score. Analyzed feature importances. Selected Logistic Regression as the preferred model for this use case.
Weeks 6–7	Reporting	Compiled all findings into this written report. Documented methodology, results, limitations, and recommendations.
Weeks 7–8	Presentation	Reviewed the full project and prepared a video presentation of findings.

X. CONCLUSION

The Logistic Regression model emerged as the more effective choice for this classification problem, achieving an AUC of 0.8374 and a recall of 76.60% for the attrition class on the test set. These results indicate that the model successfully identified approximately three out of every four employees who had left — a meaningful capability for a proactive retention system. Monthly income, overtime, age, total working years, and years with the current manager were identified as the most influential predictors of attrition, pointing to specific areas where organizations can take targeted action.

Several limitations should be acknowledged. The IBM dataset is synthetic, meaning that models trained on it may not directly transfer to real organizational data without retraining. The dataset also reflects a single point in time, capturing no longitudinal changes in employee behavior or organizational conditions.

Future work could address these limitations in several ways. Integrating real employee data through anonymized, consent-based partnerships with organizations would strengthen model generalizability. Incorporating natural language processing on exit interview data or engagement survey responses — an approach explored by Lee and Algarra [11] — could add interpretive depth beyond structured HR variables. Developing a real-time scoring system that flags individual employees for HR review, combined with periodic retraining as organizational conditions change, could transform this analytical framework into a deployed, living tool for workforce stability management.

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